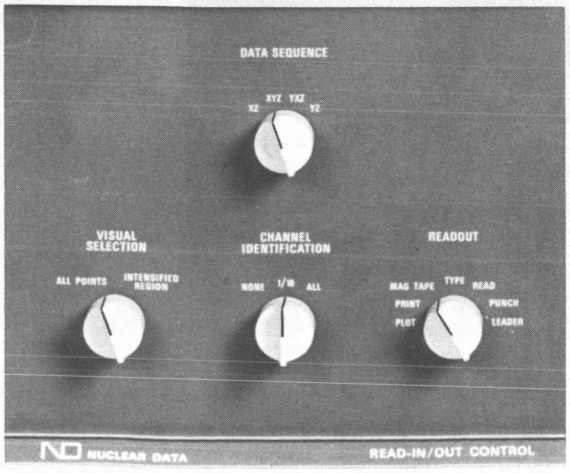
TECHNICAL SPECIFICATIONS

NUCLEAR DATA INC.
100 West Golf Road
Palatine, Illinois 60067
Phone A.C. 312 • 529-4600



SERIES 3300 READ-IN/OUT CONTROL

The Read-In/Out Control Module provides control facilities for readout of stored data from the System Memory via a wide variety of analog and digital output devices. Included are facilities for high speed analog recorders, the IBM Model B Output Typewriter, the ND-316 Autofinger/IBM 721 Selectric Typewriter combination, Tally Paper Tape Perforator, and, upon request, high speed parallel printers. High speed magnetic tape readout of arithmetic data (counts) via an external Magnetic Tape Control Unit and tape transport can also be selected.

Control facilities for read-in of data via either a Tally Paper Tape Reader or an external Magnetic Tape Control Unit and tape transport are also included. In addition, front panel controls are provided for selection of address identification and arithmetic data sequence during digital readout.

Punched paper tape readout format conforms to the American Standard Code for Information Interchange, unless otherwise specified at time of purchase. Arithmetic data is punched in 1-2-4-8 binary-coded decimal form. An internal switch enables selection of either ASCII Code or IBM Code for punched paper tape read-in.

In conjunction with the Master Control Module and the Analog Display Configuration Module, a front panel control enables data to be visually selected for readout. The visual selection is accomplished by either digital discrimination with the Master Control Module, or by analog discrimination with the Analog Display Configuration Module.

Analog Readout Speed

Determined by maximum rate of plotter (typically 20 points per second for Houston 6550).

Digital Readout Speed

IBM Model B Typewriter: 10 characters per second; non-significant zeros are suppressed.

ND-316 Autofinger/IBM 721 Selectric: 12 characters per second; non-significant zeros are suppressed.

Tally Paper Tape Punch: 60 characters per second (120 characters per second upon request).

Parallel Printer: Determined by maximum rate of printer (30 lines per second for Franklin Model 1230).

Digital Read-in Speed

Tally Paper Tape Reader: 60 characters per second (120 characters per second upon request).

Address Identification

Address Identification during readout can be selected for either every tenth channel or every channel. It can also be disabled, if desired.

Data Sequence Selection

When address identification of data to be read out has been selected, the format can be chosen as follows:

XZ — Arithmetic data read out from each address location is preceded by either an absolute channel number, or the relative channel number within a selected memory group size, depending on whether single or multi submode operation has been selected at the Master Control Module.

XYZ — Arithmetic data is preceded by (1) the relative channel number within a selected memory group size (X), and (2) the consecutive number of the group (Y) being read out.

YXZ — Arithmetic data is preceded by (1) the consecutive number of the group being read out (Y) and (2) the relative channel number within a selected memory group size (X).

YZ — Arithmetic data is preceded by the consecutive number of the group being read out.

Visual Readout Selection

A facility is provided to enable readout of only that data chosen by region of interest display intensification.

Punched Tape Readout Code

Data is read out with ASCII Code, unless otherwise specified at time of purchase. Either an IBM compatible or ASCII Code can be selected for read-in of data with punched paper tape.

Type/Punch/Read Timing Generator

A timing generator is included which programs the sequence of operations for readout with either a typewriter or paper tape perforator, or read-in with a paper tape reader.

Zero Suppression Generator

Circuitry is included for suppression of non-significant zeros during typewriter readout.

B.C.D. to Decimal Decoder

Four-line B.C.D. to ten-line decimal decoder circuitry for typewriter readout is included.

Punched Paper Tape Coder

Coding circuitry is included to encode punched paper tape format during readout, and to decode punched paper tape format during read-in.

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